

SMART CAMPUS MANAGEMENT SYSTEM (SCMS)

Teach2Give Month 2 Project Proposal
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Overview

The **Smart Campus Management System (SCMS)** is a web-based platform designed to streamline and digitize university operations — from student registration and lecturer management to attendance, grades, and exam eligibility.

The system helps:

- **Students:** Track classes, assignments, and performance.
- **Lecturers:** Manage courses, grades, and availability.
- **Administrators:** Oversee exam eligibility, fee clearance, and academic progress.

This project gives trainees practical experience in backend–frontend integration, API testing, and automated QA following Agile Scrum practices.

Team Setup and Methodology

Team size: 3–4 members

Methodology: Agile Scrum

Tool: Jira (for backlog, sprint, and ticket management)

Deliverables per sprint: Defined user stories, developed features, and test coverage.

Roles

Role	Responsibilities
Product Owner	Defines project goals, manages backlog, and validates deliverables
Scrum Master	Facilitates sprints, stand-ups, and removes blockers
Developers / QA Engineers	Build modules, test APIs, and ensure overall software quality

Collaboration Tools

- **Jira** – Sprint tracking & QA testing cycles
- **GitHub** – Version control & code collaboration

- **Postman** – API validation and documentation
- **Notion / Google Docs** – Documentation and reporting

System Architecture

Backend: Node.js + Express.js

Architecture: Layered Modular (Controller → Service → Repository)

Database: Microsoft SQL Server / PostgreSQL

Frontend: React.js + TypeScript

Testing:

- Manual Testing
- Unit & Integration: Jest
- Performance: k6
- End-to-End (E2E): Cypress

Version Control: Git & GitHub

Deployment (Optional): Render / Azure / AWS / Vercel

Core Features

1. Authentication & Roles

- Secure registration and login (JWT)
- Roles: Admin, Lecturer, Student
- Role-based access to dashboards and actions

2. Student Module

- Register and manage student details
- View timetable, grades, and attendance
- Check exam eligibility based on:
 - Attendance 75%
 - Coursework average 50%
 - Fees cleared

3. Lecturer Module

- Manage assigned units and courses
- Upload grades, assignments, and CAT results
- Mark attendance per class
- Set availability for consultations

4. Course & Unit Management

- Manage courses, units, and linked lecturers
- Define timetables (day, time, venue)
- Track enrolled students per unit

5. Academic Performance & Eligibility Engine

- Auto-calculate performance
- Determine exam eligibility based on attendance, coursework, and fees
- Auto-flag ineligible students
- Admin override for exceptions

6. Attendance Tracking

- Mark attendance per class
- Generate summaries and reports
- Feed data into eligibility computations

7. Finance Module

- Record student fee payments
- Verify financial clearance before exams
- Generate clearance reports

8. Timetable & Scheduling

- Manage lecture, CAT, and exam timetables
- Notify students of changes
- Allow lecturers to view and adjust sessions

Database Design

Main Tables:

- Users
- Courses
- Units
- Enrollments
- Attendance
- Assignments
- CATs
- Grades
- Payments
- Timetable

Optional Tables: Notifications, Consultations

Backend Deliverables

Architecture:

```
/src
  /controllers    → Handles incoming API requests
  /services       → Business logic (eligibility, grading)
  /repositories   → Database operations
  /models         → Database schemas
  /middlewares    → Authentication & validation
  /routes         → API endpoint definitions
  /tests         → Unit & integration tests
```

API Endpoints:

- Authentication: Register, Login, Logout
- Student: Dashboard, grades, attendance, eligibility
- Lecturer: CATs, assignments, attendance, grades
- Admin: Users, payments, eligibility
- Timetable: View/update schedules
- Finance: Record/view payments

Frontend Deliverables

	Page	Description
Main Pages:	Login / Register	Secure login and registration
	Dashboard (Student)	Attendance, grades, eligibility
	Dashboard (Lecturer)	Manage courses and students
	Admin Panel	Manage users, payments, clearance
	Timetable Page	View lectures, CATs, exams
	Payment Page	Display and update fee records
	Eligibility Report Page	Show students cleared for exams

Expected QA / QE Practice Areas

Practice Area	Tools / Deliverables
Manual Testing	Test cases for endpoints
API Validation	Postman collections
Unit Testing	Jest for controllers & services
Integration Testing	Jest + Supertest
Performance Testing	k6 for load & stress tests
E2E Testing	Cypress for system flow
Bug Tracking	Jira (Sprint Review, QA Reports)
Documentation	Test cases, coverage, QA summary

Summary

Aspect	Description
System Type	Web-based university management platform
Stakeholders	Students, Lecturers, Admins
Architecture	Node.js + Express + SQL + React
Goal	Automate and simplify academic operations
Unique Feature	Automated Exam Eligibility Engine
Development Duration	1 Month (MVP)
Scalability	Expandable into full ERP (hostel, library, etc.)